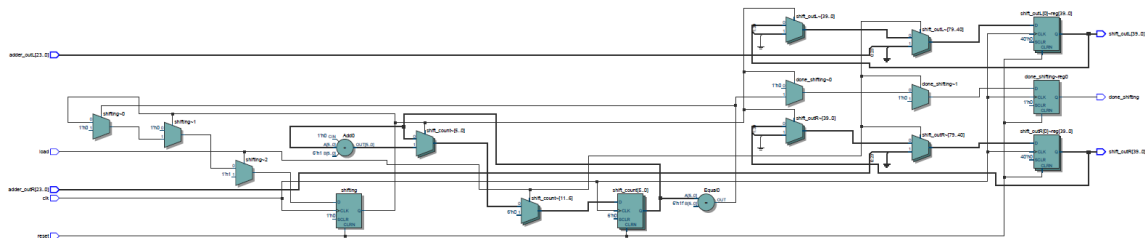


ShiftRightRegister

This is a 1 bit shifter, it takes the output of each side of the adder and shifts it to the right by 1-bit, and sends the results back to the adder where the result will be accumulated with the new datain sample. This process repeats until the convolution is completed, then the final result is sent to the PISO block.

Block Diagram of ShiftRightRegister:



WaveView of the Simulated Testbench

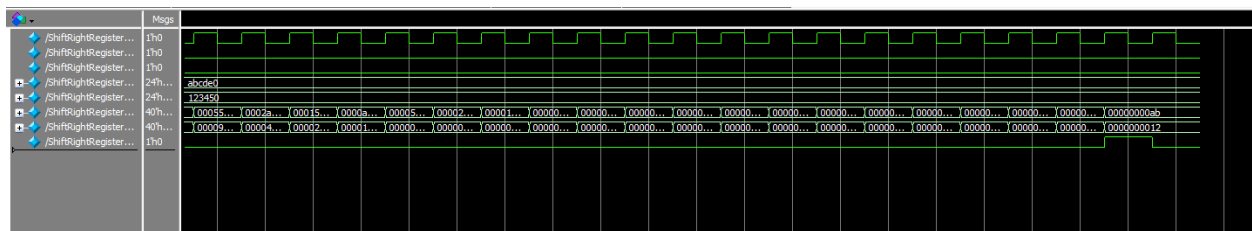


Table showing the simulated result

```
# run -all
# T=0 | load=0 | done=0 | shift_outL=0x0000000000 | shift_outR=0x0000000000
# T=20000 | load=1 | done=0 | shift_outL=0x0000000000 | shift_outR=0x0000000000
# T=25000 | load=1 | done=0 | shift_outL=0xabcd000000 | shift_outR=0x1234500000
# T=30000 | load=0 | done=0 | shift_outL=0xabcd000000 | shift_outR=0x1234500000
# T=35000 | load=0 | done=0 | shift_outL=0x55e6f00000 | shift_outR=0x091a280000
# T=40000 | load=0 | done=0 | shift_outL=0x2af3780000 | shift_outR=0x048d140000
# T=45000 | load=0 | done=0 | shift_outL=0x1579bc0000 | shift_outR=0x002468a000
# T=50000 | load=0 | done=0 | shift_outL=0xabcd000000 | shift_outR=0x0123450000
# T=55000 | load=0 | done=0 | shift_outL=0x055e6f0000 | shift_outR=0x0091a28000
# T=60000 | load=0 | done=0 | shift_outL=0x02af378000 | shift_outR=0x0048d14000
# T=65000 | load=0 | done=0 | shift_outL=0x01579bc000 | shift_outR=0x002468a000
# T=70000 | load=0 | done=0 | shift_outL=0x0000abcd00 | shift_outR=0x0001234500
# T=75000 | load=0 | done=0 | shift_outL=0x000055e6f0 | shift_outR=0x0000091a28
# T=80000 | load=0 | done=0 | shift_outL=0x00002af378 | shift_outR=0x0000048d14
# T=85000 | load=0 | done=0 | shift_outL=0x00001579bc | shift_outR=0x000002468a
# T=90000 | load=0 | done=0 | shift_outL=0x000000abcd | shift_outR=0x0000001234
# T=95000 | load=0 | done=0 | shift_outL=0x00000055e6 | shift_outR=0x000000091a
# T=100000 | load=0 | done=0 | shift_outL=0x0000002af3 | shift_outR=0x000000048d
# T=105000 | load=0 | done=0 | shift_outL=0x0000001579 | shift_outR=0x0000000246
# T=110000 | load=0 | done=0 | shift_outL=0x0000000000 | shift_outR=0x0000000000
# T=115000 | load=0 | done=0 | shift_outL=0x0000000000 | shift_outR=0x0000000000
# T=120000 | load=0 | done=0 | shift_outL=0x0000000000 | shift_outR=0x0000000000
# T=125000 | load=0 | done=0 | shift_outL=0x0000000000 | shift_outR=0x0000000000
# T=130000 | load=0 | done=0 | shift_outL=0x0000000000 | shift_outR=0x0000000000
# T=135000 | load=0 | done=0 | shift_outL=0x0000000000 | shift_outR=0x0000000000
# T=140000 | load=0 | done=0 | shift_outL=0x0000000000 | shift_outR=0x0000000000
# T=145000 | load=0 | done=0 | shift_outL=0x0000000000 | shift_outR=0x0000000000
# T=150000 | load=0 | done=0 | shift_outL=0x0000000000 | shift_outR=0x0000000000
# T=155000 | load=0 | done=0 | shift_outL=0x0000000000 | shift_outR=0x0000000000
# T=160000 | load=0 | done=0 | shift_outL=0x0000000000 | shift_outR=0x0000000000
# T=165000 | load=0 | done=0 | shift_outL=0x0000000000 | shift_outR=0x0000000000
# T=170000 | load=0 | done=0 | shift_outL=0x0000000000 | shift_outR=0x0000000000
# T=175000 | load=0 | done=0 | shift_outL=0x0000000000 | shift_outR=0x0000000000
# T=180000 | load=0 | done=0 | shift_outL=0x0000000000 | shift_outR=0x0000000000
# T=185000 | load=0 | done=0 | shift_outL=0x0000000000 | shift_outR=0x0000000000
# T=190000 | load=0 | done=0 | shift_outL=0x0000000000 | shift_outR=0x0000000000
# T=195000 | load=0 | done=0 | shift_outL=0x0000000000 | shift_outR=0x0000000000
# T=200000 | load=0 | done=0 | shift_outL=0x0000000000 | shift_outR=0x0000000000
# T=205000 | load=0 | done=0 | shift_outL=0x0000000000 | shift_outR=0x0000000000
# T=210000 | load=0 | done=0 | shift_outL=0x0000000000 | shift_outR=0x0000000000
# T=215000 | load=0 | done=0 | shift_outL=0x0000000000 | shift_outR=0x0000000000
# T=220000 | load=0 | done=0 | shift_outL=0x0000000000 | shift_outR=0x0000000000
# T=225000 | load=0 | done=0 | shift_outL=0x0000000000 | shift_outR=0x0000000000
# T=230000 | load=0 | done=0 | shift_outL=0x0000000000 | shift_outR=0x0000000000
# T=235000 | load=0 | done=0 | shift_outL=0x0000000000 | shift_outR=0x0000000000
# T=240000 | load=0 | done=0 | shift_outL=0x0000000000 | shift_outR=0x0000000000
# T=245000 | load=0 | done=0 | shift_outL=0x0000000000 | shift_outR=0x0000000000
# T=250000 | load=0 | done=0 | shift_outL=0x0000000000 | shift_outR=0x0000000000
# T=255000 | load=0 | done=0 | shift_outL=0x0000000000 | shift_outR=0x0000000000
# T=260000 | load=0 | done=0 | shift_outL=0x0000000000 | shift_outR=0x0000000000
# T=265000 | load=0 | done=0 | shift_outL=0x0000000000 | shift_outR=0x0000000000
# T=270000 | load=0 | done=0 | shift_outL=0x0000000000 | shift_outR=0x0000000000
# T=275000 | load=0 | done=0 | shift_outL=0x0000000000 | shift_outR=0x0000000000
# T=280000 | load=0 | done=0 | shift_outL=0x0000000000 | shift_outR=0x0000000000
# T=285000 | load=0 | done=0 | shift_outL=0x0000000000 | shift_outR=0x0000000000
# T=290000 | load=0 | done=0 | shift_outL=0x0000000000 | shift_outR=0x0000000000
# T=295000 | load=0 | done=0 | shift_outL=0x0000000000 | shift_outR=0x0000000000
# T=300000 | load=0 | done=0 | shift_outL=0x0000000000 | shift_outR=0x0000000000
# T=305000 | load=0 | done=0 | shift_outL=0x0000000000 | shift_outR=0x0000000000
# T=310000 | load=0 | done=0 | shift_outL=0x0000000000 | shift_outR=0x0000000000
# T=315000 | load=0 | done=0 | shift_outL=0x0000000000 | shift_outR=0x0000000000
# T=320000 | load=0 | done=0 | shift_outL=0x0000000000 | shift_outR=0x0000000000
# T=325000 | load=0 | done=0 | shift_outL=0x0000000000 | shift_outR=0x0000000000
# T=330000 | load=0 | done=0 | shift_outL=0x0000000000 | shift_outR=0x0000000000
# Shifting done at time 345000
# Final shift_outL = 0x0000000000
# Final shift_outR = 0x0000000000
# T=345000 | load=0 | done=1 | shift_outL=0x0000000000 | shift_outR=0x0000000000
# T=350000 | load=0 | done=0 | shift_outL=0x0000000000 | shift_outR=0x0000000000
# ** Note: $finish : C:/Users/lanjm/OneDrive/Documents/Grad School/UTD/Spring 2025/CE 6306 ASIC Design/Projects/Midterm/ALU_Block/ShiftRightRegister/ShiftRightRegister_tb.v(49)
# Time: 365 ns Iteration: 0 Instance: /ShiftRightRegister_cb
```